

Operations Research for Health Care

Lesson 6/15

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Outline of this lecture

Hierarchical health care location problems

Review and classification

Optimal coverage radius for each level

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Review of hierarchical health care location [Şahin and Süral, 2007] and [Contreras and Ortiz-Astorquiza, 2019]

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- ▶ **Flow pattern**: single-flow, multi-flow, bidirectional-flow. (referral system).
- ▶ **Service**: successively inclusive and exclusive (nested and non-nested).
- ▶ **Spatial configuration**: Coherent or non-coherent system.
- ▶ **Objectives**: minimizing total transportation costs/distances (*median*), cost for opening (*fixed charge*), maximizing coverage (*maximal covering*), other.

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Given a set of patients that demand one or k services and a set of facilities associated with k services, select a set of facilities to open so that each patient receives the requested service(s), while optimizing a criteria.

Flow pattern [Şahin and Süral, 2007] and [Contreras and Ortiz-Astorquiza, 2019]

- ▶ **Single-flow**: starts from (a demand point) **level 0** visit facilities in each level until reaching the highest **level k**. *In a referral system*, a proportion of patients served at each level are referred to higher levels.

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- ▶ **Bidirectional-flow**: Demand flow routed in both directions. Flow starts at the (demand point) patient level and visit some or all higher-levels and then is routed back to lower-levels until it arrives to a demand point.

Regardless of the flow pattern, a portion of patients demand served at a given level is eventually *referred* for additional services to a higher level. This is denoted as **referral systems**.

Service and Spatial configuration [Şahin and Süral, 2007] and

[Contreras and Ortiz-Astorquiza, 2019]

Service hierarchy:

- ▶ **Successively inclusive (nested)**: a higher-level facility **provides all the services** provided by a lower level facility **and** at least *one additional service*.
- ▶ **Successively exclusive, non-nested**: facilities at each level offer different services.

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Spatial configuration:

- ▶ **Coherent system** (single assignment): all demand sites that are assigned to a particular lower-level facility are assigned to one and the same higher-level facility, i.e., all demand flow entering to a particular facility at a lower level **is assigned to exactly one** facility at a higher-level.

Decisions:

- ▶ **Design**: Facility location and edge activation (**where** to open the facilities).

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- ▶ **Design:** Facility location and edge activation (**where** to open the facilities).
- ▶ **Tactical:** Allocation (which facilities will **serve** each demand)

Health-care systems are one of the most studied systems in the literature.

- ▶ 1970: Schultz: Multi-flow, nested, maximize benefit.
- ▶ 1973: Calvo, Marks: Multi-flow, nested, multi-objective.
- ▶ 1977: Dökmeci: Multi-flow, nested, level-by-level, fixed-charge.
- ▶ 1979: Dökmeci: Multi-flow, nested, level-by-level, bi-criteria.
- ▶ 1979: Narula, Ogbu: Multi-flow, nested, coherent referral system, median.
- ▶ 1980: Parr: Multi-flow, nested, maximize benefit, parametric.
- ▶ 1983: Tien, El-Tell, Simons: Multi-flow, median.
- ▶ 1984: Tien, El-Tell: Single-flow, coherent region Jordan, median.
- ▶ 1985: Narula, Ogbu: Multi-flow, nested, coherent referral, methods.
- ▶ 1987: Church, Eaton: Multi-flow, nested, referral system, maximal covering.
- ▶ 1993: Serra, ReVelle: Multi-flow, nested, coherent referral system, median.

Hierarchical health care papers [Şahin and Süral, 2007]

- ▶ 1994: Gerrard, Church: Multi-flow, nested, referral at Uganda, max coverage.
- ▶ 1994: Serra, ReVelle: Multi-flow, nested, coherent referral, median, methods.
- ▶ 1996: Serra: Multi-flow, nested, coherent referral system, bi-objective.
- ▶ 1997: Okabe, Okunuki, Suzuki: Multi-flow, nested, budged constrained, median.
- ▶ 2001: Sahin, Meral: Multi-flow, nested, coherent referral Turkey, median.
- ▶ 2002: Galvao, Espejo, Boffey: Multi-flow, nested, flow referral system, median.
- ▶ 2002: Sahin: Multi-flow, nested, coherent referral system.
- ▶ 2003: Boffey, Yates, Galvao: Multi-flow, nested, coherent referral system.
- ▶ 2003: Espejo, Galvao, Boffey: Multi-flow, nested, maximal coverage.

Review of hierarchical health care location [Şahin and Süral, 2007]

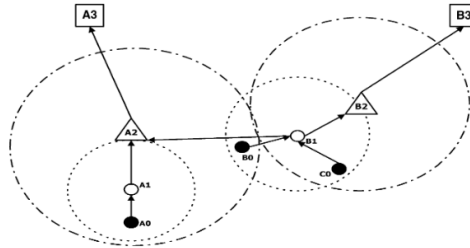


Fig. 1. A single-flow, non-nested and non-coherent structure. A hierarchical system with three levels is shown. Sites are marked with letters and numbers denoting the level. Service areas of different-level facilities are circled with different patterns. A customer to be served by the highest-level facility goes to a lowest-level facility first and then passes through all levels until the top. As the facilities are non-nested, different-level facilities are marked with different shapes, denoting that different services are provided at different levels: a level 1 facility is marked with a white circle, a level 2 facility is marked with a triangle, a level 3 facility is marked with a square and dark circles represent the customer sites. In a non-coherent structure, the customers assigned to the same facility at the lowest level may be assigned to different facilities at a higher level: B0 is serviced by B1 for its level 1 demand. Although B1 works with B2 for its level 2 demand, B0 is serviced by A2 for its level 2 demand.

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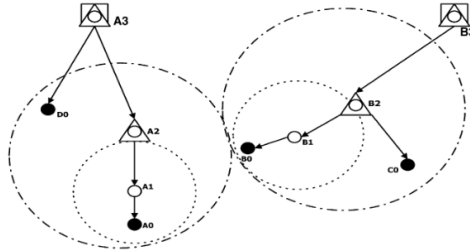


Fig. 2. A multi-flow, nested and coherent structure. A three-level system is presented. It is a multi-flow system as goods are shipped from a higher-level facility to any lower-level facility (see the flows from A3 to D0 and B2 to C0). Since the facilities are nested (i.e. all services at lower levels are available at higher levels), a higher-level facility is denoted by all shapes of lower-level facilities (for example, a level 3 facility is marked with a square, a triangle and a circle). As the structure is coherent, a lower-level service area is a subset of a higher-level service area.

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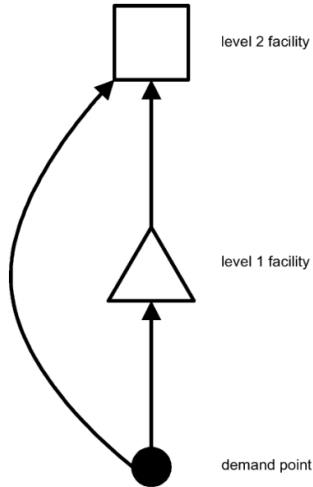


Fig. 3. A schematic representation of flow patterns. In a single-flow system, flow starts from demand site, passes a level 1 facility and ends at a level 2 facility. In a multi-flow system, demand can be served by a level 2 facility for its both level 1 and 2 demands. If facilities are capacitated, a portion of level 1 demand can be served by a level 1 facility and the remaining portion by a level 2 facility. However, no such fractional assignments are observed in the uncapacitated version of the problem: demand is single-sourced by a facility either at level 1 or 2 for its level 1 demand, and served by a level 2 facility either directly or via a level 1 facility for its level 2 demand.

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Hierarchical health care location problems

Review and classification

Optimal coverage radius for each level

Review and classification [Narula, 1986]

A survey of minimum k -levels ($k \geq 2$) hierarchical location-allocation problems for health care planning.

(1) Health center - basic health care, (3) hospitals - medium and high complexity procedures, and (2) medical center - more services and equipment to treat all cases.

- ▶ **The goal:** locate p_1 level 1 facilities, ..., p_k level k facilities among n potential locations, such that $n \geq \sum_{i \in 1..k} p_i$, and allocating demand points to selected facilities.
- ▶ It is a generalization of p -median (minimizing total weighted distance) or p -center problem (minimizing maximum distance, covering problem). An optimum solution to the problem occurs at the vertices of the network (Hakimi)[20].
- ▶ *Intermediate facility location problems:* demand location and primary facilities are fixed and known. The objective is to locate only intermediate facilities [26,30,31,50].

Review and classification [Narula, 1986]

Definitions: p_i facilities, $i = 1, \dots, k$, among $n \geq \sum_{i \in 1..k} p_i * w_j$ demand, $j = 1, \dots, n$, minimizing total weighted distance.

Successively inclusive facility hierarchy: In a health care system, a facility at level m ($m = 1, \dots, k$) offers services of type $1, \dots, m$.

Locally inclusive service hierarchy: If a level m facility, $m = 1, \dots, k$, at a location i offers type $1, \dots, m$ services only to location i and only type m services to a location j , $j \neq i$. E.g.:

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- ▶ **level 1:** basic care (1)
- ▶ **level 2:** basic (1) and medium complexity (2) care (*only to location 2*). Medium complexity care is offered to location 1, **but basic care is not offered to location 1.**

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- ▶ **level 3:** basic (1), medium (2) and high complexity (3) care (*only to location 3*). High complexity care is offered for locations 2, and 1, **but basic and medium complexity care are not offered for locations 1 and 2**.

Review and classification [Narula, 1986]

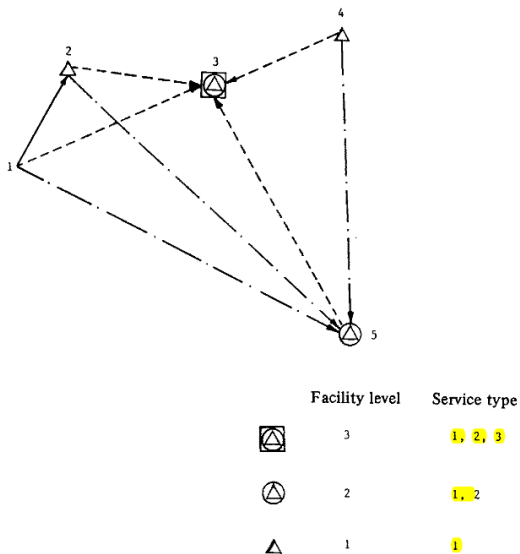


Fig. 1. An example of locally inclusive service hierarchy.

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Successively exclusive facility hierarchy, a facility level m ($m = 1, \dots, k$) offers services **unique to it**, i.e. a facility at level m offers services of type m only, e.g. production-distribution system. E.g:

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- ▶ **level 2:** service level 2 (to locations 1,2,3).

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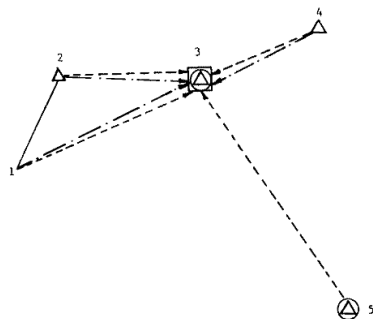
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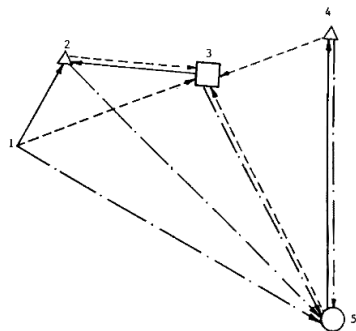
- ▶ **level 1:** service level **1** (to locations 1,2,3)
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- ▶ **level 3:** service level **3** (to locations 1,2,3).

Review and classification [Narula, 1986]



	Facility level	Service type
	3	1, 2, 3
	2	1, 2
	1	1

Fig. 2. An example of successively **inclusive** service hierarchy.



	Facility level	Service type
	3	3
	2	2
	1	1

Fig. 3. An example of successively **exclusive** service hierarchy.

Review and classification [Narula, 1986]

- ▶ Calvo and Marks (1973)[4] described a (locally inclusive) 3-hierarchical health care delivery system in detail and developed an analytical model for a k -hierarchical system. *They assumed that the patients can be divided into k groups according to their health care needs.* This model could not be solved directly, therefore, Tien et al.(1983) [45] corrected two typographical errors, formulated and solved a successfully inclusive model.
- ▶ Narula and Ogbu (1979)[34] assumed that the *patients really do not know the type of service they need*, therefore, they formulated a 2-hierarchical MILP problem assuming that θ proportion of the patients arriving at a health center(on level 1) will be referred to a hospital (on level 2).

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The logic behind health service coverage [Schultz, 1970]

A logical model for the planning of health care facilities provides a method is for determining the **optimal service area** and an approach for generating a socially optimal health care facility pattern for a (simplified) region.* The degree of decentralization:

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- ▶ the *frequency* with which people in a region use health care facilities; and
- ▶ external social costs: *increased travel and reduced utilization of facilities*.

A “*socially optimal*” solution can be obtained by:

minimum cost , assuming fixed output requirements;

maximum net benefit (benefit minus cost) without constraints (*this paper**);

maximum net benefit subject to output, input and budget constraints.

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- ▶ **Net social benefit = demand for services, the unit cost of services, travel cost.**

ASSUMPTIONS: The region has a homogeneous population with a uniform areal distribution. The facility hierarchy is *successively inclusive* and consists of 3 levels:

- ▶ **Type 1:** Neighborhood health center. *Order 1* services: Primary diagnosis and outpatient treatment. Minimal short term inpatient care.
- ▶ **Type 2:** General hospital. *Order 2* services: Specialized diagnosis and outpatient treatment. Short and long term inpatient care.
- ▶ **Type 3:** Medical center *Order 3* services: Complex diagnosis and outpatient treatment. Intensive short and long term inpatient care.

ASSUMPTIONS:

- ▶ Economies of scale affect the cost of production of services but urbanization and localization economies are negligible;
- ▶ The distance-demand function for services varies among orders but is not a function of price or available capacity;
- ▶ Each consumer of a health service uses the nearest facility from which it is available;
- ▶ The cost of travel between points is proportional to airline distance;
- ▶ The cost of travel is the same for all users of an order of services.
- ▶ Social benefits/unit of service vary with the order and utilization level in a subarea but are the same for all population subgroups.

Service costs: sum of **capital** and **operating costs/service** unit (sum of inpatient, outpatient and emergency cases.) Let k be the order of services, p_k is the average cost/unit of service, Y_k is the services provided/day; $C1_k$ is the average capital cost/unit of service; $C2_k$ is the average operating cost/unit of service.

$$p_k(Y_k) = C1_k(Y_k) + C2_k(Y_k) \quad \forall k \in K$$

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Distance-demand: Each order of services has a (negative power $z ax^{-b}$, or negative exponential $z e^{-bx+a}$) distance-demand function (USA,FR empirical studies [5,10]).

$$z_k(x) = a_k * e^{-\alpha_k * x_k} \quad \forall k \in K$$

Where z_k is the utilization, i.e. the number of service units demanded/1000 population in a subarea of the region; x_k is the distance of the subarea from a facility; a_k is the utilization for families near the facility; α_k is the rate at which utilization decays with distance.

- ▶ **Type 1:** *Neighborhood health center*: The cost of using the facility will often be greater than the expected private benefits to users and demand for first order services will decrease rapidly with distance.
- ▶ **Type 3:** *Medical center*: people referred to the third order services unique to a medical center will be those requiring services which are imperative. The cost of travel to them will be small compared with the expected benefits so that the distance-demand for third order services will be rather level

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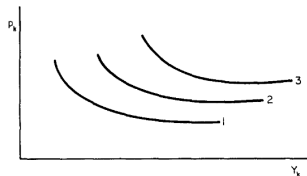


FIG. 1. Service cost functions for three orders of services.

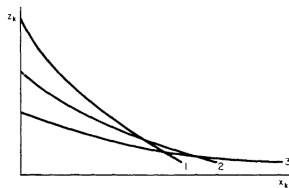


FIG. 2. Distance-demand functions for three orders of services.

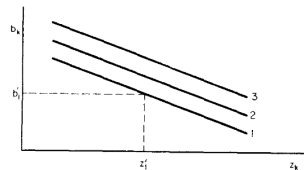


FIG. 3. Social benefit functions for three orders of services.

The logic behind health service coverage [Schultz, 1970]

Travel cost: Cost of transportation, time and inconvenience/mile between the facility and a patient's residence.

$$c_a k \quad \forall k \in K$$

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Social benefit: The present value of future private benefits and external effects derived from different levels of utilization of the facility by residents. When utilization is low, a large proportion of the service units consumed have high social benefits. When utilization is relatively high, more low benefit services are being used.

$$b_k(z_k) = b'_k + \beta_k(z'_k - z_k) \quad \forall k \in K$$

where b_k is the estimated social benefit, in dollars, per service unit consumed; z_k is the utilization in a subarea; z'_k is the “standard utilization level” for services of order k ; b'_k is the average social benefit/service unit at the standard level of utilization; β_k is the rate at which the average social benefit changes as utilization varies from the standard level.

OPTIMIZATION [Schultz, 1970]

Let:

- ▶ r : the **radius** of the service area;
- ▶ $B(r)$: the *per capita gross benefit* for a service area *with radius r* ;
- ▶ $C(r)$: the *per capita cost* for the *same area*.

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The **maximization** of net benefit/*capita* for the population served is:

$$r^* = \max_r [B(r) - C(r)]$$

First order conditions: $d/dr[B(r) - C(r)] = 0$ is equivalent to having the marginal per capita gross benefit (with respect to the service radius) equal the marginal per capita cost: $dB/dr = dC/dr$

OPTIMIZATION [Schultz, 1970]

Optimal radius: r^* . Let x be the distance from the facility to the ring, and w be the uniform population density (1000 person/mile). The demand in a ring of unit width centered on a facility is:

$$y(x) = 2\pi xwz(x)$$
$$Y(r) = \int_0^r y(x)dx$$

Gross benefit/capita is the sum of the benefits divided by the population of the area.

$$B(r) = \int_0^r y(x)b[z(x)]dx/\pi r^{2w}$$

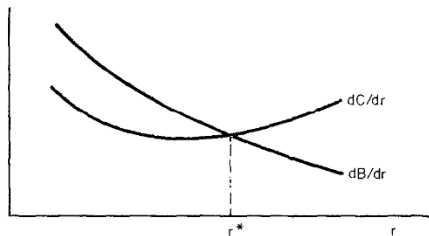
Cost/capita is the total service + travel costs divided by the population of the area.

$$C(r) = [p[Y(r)]Y(r) + \int_0^r y(x)c3xdx]/\pi r^{2w}$$

OPTIMIZATION [Schultz, 1970]

Costs: “U-shaped”: beyond some radius, travel cost rise more rapidly than service price falls.

Benefit: Downward sloping if utilization decreases more rapidly than the benefit/service unit increases.



Marginal cost and benefit functions for a single order of services.

Optimal scale: Since each facility must meet demand, we can find the optimal scale for each order of services it offers by computing: $Y^* = Y(r^*)$.

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